

Interleaving structure and mod-2 pattern laws for the pendant-ladder matching-count recurrences (Lean 4 machine-checked claim of record)

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Abstract

Let $a(n)$ be the number of matchings (independent edge sets, the empty one included) of the ladder graph $P_n \square K_2$ with one pendant leaf attached at every 1-based odd column on a uniform rail, starting 3, 10, 46, 141, 660, 2015, $\dots$. Three integer sequences are defined by literal seeds and linear recurrences: one by eight seeds and the even-lag order-8 recurrence $x(n) = 16x(n-2) - 25x(n-4) + 10x(n-6) - x(n-8)$, and two subsequences by four seeds each under the shared order-4 recurrence with characteristic polynomial $x^4 - 16x^3 + 25x^2 - 10x + 1$. We state and machine-check in Lean 4 (v4.34.0, mathlib v4.34.0) four laws for these recurrence-defined sequences: the interleaving identity between the full sequence and the two subsequences; an explicit period-12 mod-2 pattern of the full sequence; oddness of the odd-position subsequence exactly at $k \equiv 0, 4 \pmod{6}$; and evenness of the even-position subsequence exactly at $k \equiv 0 \pmod{3}$. Every proof compiles with no `sorry`, and `#print axioms` reports only {propext, Quot.sound}. The odd-position values coincide with OEIS A386889 (Dresden and Demirkol, 2025-09-04), an occupied record; the novelty asserted here is limited to the full order-8 recurrence, the even-position subsequence, the interleaving identity and the mod-2 patterns. The combinatorial identification of the sequences with matching counts is not kernel-proved and is stated as a conjecture. This note is a priority claim of record, not a full paper.

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1 Introduction

Four theorems about three integer sequences defined by initial values and linear recurrences are reported here; conjecturally the sequences count matchings of a pendant-leaf ladder graph [4], and the recurrences were obtained by exact rational Berlekamp–Massey fitting of enumerated data, then independently recomputed [3]. The proofs were produced by the AI4Math pipeline, in which LLM workers generate statements and proof scripts and the Lean 4 kernel is the sole adjudicator at every gate: statements are frozen by hash before proving, and a separate audit re-computes the symmetric difference, scans the whole file for `sorry/admit` including comments, and re-prints the axiom dependencies [1, 2]. Verification strength is stated below verbatim. This note is a priority claim of record deposited so that a dated, citable public record exists ahead of any narrative paper; a fuller paper for the same result is prepared separately and will reference this deposit. No literature review is attempted here.

2 Statement

The three sequences are *defined* by the displayed initial values and recurrences; 0-based indexing with `apend` $n = a(n + 1)$, `opend` $k = a(2k + 1)$ (odd-position side) and `epend` $k = a(2k + 2)$ (even-position side). Occupancy fence, carried verbatim from the audit record: the numerical values of `opend` coincide with OEIS **A386889** [6] (Dresden and Demirkol, 2025-09-04; occupied, values cross-checked to $k = 15$); the novelty of this line is strictly limited to the full order-8 recurrence plus the even-position subsequence (`epend` side); no “new sequence” claim about the whole family is made anywhere in this note.

Theorem 1 (interleaving identity). *For the sequences of Listing 1 and every $k \in \mathbb{N}$, `apend`($2k$) = `opend`(k) and `apend`($2k + 1$) = `epend`(k): the full order-8 recurrence is exactly the interleaving of two order-4 subsequences sharing the characteristic polynomial $x^4 - 16x^3 + 25x^2 - 10x + 1$.*

Listing 1: Frozen formal statements, verbatim lines 27–68 of `formalized/01-pend-statements.lean` (sha256 78bccc97): the three sequence definitions and Theorem 1.

```
1 /-- **序列 apend (主列, order-8 仅偶滞后递推) **.
2 初值  $a(0..7) = 3, 10, 46, 141, 660, 2015, 9440, 28814$ ;
```

```

3  递推  $a(n+8) = 16 \cdot a(n+6) - 25 \cdot a(n+4) + 10 \cdot a(n+2) - a(n)$  ( $n \geq 0$ )。
   -/
4  def append :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
5    | 0 => 3
6    | 1 => 10
7    | 2 => 46
8    | 3 => 141
9    | 4 => 660
10   | 5 => 2015
11   | 6 => 9440
12   | 7 => 28814
13   | n + 8 =>
14     16 * append (n + 6) - 25 * append (n + 4) + 10 * append (n + 2)
   - append n
15
16  /-- **序列 opend (奇位侧子列, 四阶递推) **。
17  初值  $o(0..3) = 3, 46, 660, 9440$  (= append 在下标  $0, 2, 4, 6$  处之值;
18  数值与 OEIS A386889 一致——该侧已占位, 见文件头围栏);
19  递推  $o(k+4) = 16 \cdot o(k+3) - 25 \cdot o(k+2) + 10 \cdot o(k+1) - o(k)$  ( $k \geq 0$ )。
   -/
20  def opend :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
21    | 0 => 3
22    | 1 => 46
23    | 2 => 660
24    | 3 => 9440
25    | k + 4 =>
26     16 * opend (k + 3) - 25 * opend (k + 2) + 10 * opend (k + 1)
   - opend k
27
28  /-- **序列 epend (偶位侧子列, 四阶递推) **。
29  初值  $e(0..3) = 10, 141, 2015, 28814$  (= append 在下标  $1, 3, 5, 7$  处之
   值);
30  递推  $e(k+4) = 16 \cdot e(k+3) - 25 \cdot e(k+2) + 10 \cdot e(k+1) - e(k)$  ( $k \geq 0$ )。
   -/
31  def epend :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
32    | 0 => 10
33    | 1 => 141
34    | 2 => 2015
35    | 3 => 28814
36    | k + 4 =>
37     16 * epend (k + 3) - 25 * epend (k + 2) + 10 * epend (k + 1)
   - epend k
38
39  /-- **T1 (主攻) **: 主列 order-8 递推 = 两条同系数四阶子列的交织——
40  对一切  $k$ ,  $append(2k) = opend\ k$  且  $append(2k+1) = epend\ k$ 。 -/

```

```

41 theorem apend_interleave (k : ℕ) :
42   apend (2 * k) = opend k ∧ apend (2 * k + 1) = epend k := by

```

Theorem 2 (period-12 mod-2 pattern of the full sequence). *For every $n \in \mathbb{N}$, $\text{apend}(n) \bmod 2$ depends only on $n \bmod 12$: it is 1 exactly for $n \bmod 12 \in \{0, 3, 5, 8, 9, 11\}$ and 0 otherwise.*

Listing 2: Frozen formal statement, verbatim lines 71–89 of formalized/01-pend-statements.lean: the pattern pat2 and Theorem 2.

```

1  /-- **余数样式 pat2** (12 留数类  $\mapsto$  余数) :
2  0 $\mapsto$ 1, 1 $\mapsto$ 0, 2 $\mapsto$ 0, 3 $\mapsto$ 1, 4 $\mapsto$ 0, 5 $\mapsto$ 1, 6 $\mapsto$ 0, 7 $\mapsto$ 0, 8 $\mapsto$ 1, 9 $\mapsto$ 1, 10 $\mapsto$ 0,
3    11 $\mapsto$ 1. -/
4  def pat2 : ℕ  $\rightarrow$  ℤ
5    | 0 => 1
6    | 1 => 0
7    | 2 => 0
8    | 3 => 1
9    | 4 => 0
10   | 5 => 1
11   | 6 => 0
12   | 7 => 0
13   | 8 => 1
14   | 9 => 1
15   | 10 => 0
16   | _ => 1
17  /-- **T2 (伴随) **: apend 的 mod 2 余数仅依赖  $n \bmod 12$ , 样式为 pat2
18    。 -/
19  theorem apend_mod2_period12 (n r : ℕ) (hr : n % 12 = r) :
20    apend n % 2 = pat2 r := by

```

Theorem 3 (oddness pattern of the odd-position subsequence). *For every $k \in \mathbb{N}$, $\text{opend}(k)$ is odd if and only if $k \equiv 0$ or $4 \pmod{6}$.*

Listing 3: Frozen formal statements, verbatim lines 92–106 of formalized/01-pend-statements.lean: the patterns pat6 and pat3 and Theorem 3.

```

1  /-- **余数样式 pat6**: 0 $\mapsto$ 1, 4 $\mapsto$ 1, 其余 $\mapsto$ 0 (订正后口径: 奇  $\iff k \equiv$ 
2    0 或 4 (mod 6)). -/

```

```

2 def pat6 : ℕ → ℤ
3   | 0 => 1
4   | 4 => 1
5   | _ => 0
6
7 /-- **余数样式 pat3**: 0→0, 其余→1 (订正后口径: 偶  $\iff k \equiv 0 \pmod{3}$ )。 -/
8 def pat3 : ℕ → ℤ
9   | 0 => 0
10  | _ => 1
11
12 /-- **T3 之 opend (伴随)**: opend k 的 mod 2 余数仅依赖 k mod 6,
    样式为 pat6
13 (即 opend k 为奇  $\iff k \equiv 0 \text{ 或 } 4 \pmod{6}$ )。 -/
14 theorem opend_mod2_period6 (k r : ℕ) (hr : k % 6 = r) :
15   opend k % 2 = pat6 r := by

```

Theorem 4 (evenness pattern of the even-position subsequence). *For every $k \in \mathbb{N}$, $\text{opend}(k)$ is even if and only if $k \equiv 0 \pmod{3}$.*

Listing 4: Frozen formal statement, verbatim lines 109–112 of formalized/01-pend-statements.lean: Theorem 4.

```

1 /-- **T3 之 epend (伴随)**: epend k 的 mod 2 余数仅依赖 k mod 3,
    样式为 pat3
2 (即 epend k 为偶  $\iff k \equiv 0 \pmod{3}$ )。 -/
3 theorem epend_mod2_period3 (k r : ℕ) (hr : k % 3 = r) :
4   epend k % 2 = pat3 r := by

```

Conjecture 1 (combinatorial bridge, not proved here). For all $n \geq 0$, $\text{apend}(n)$ equals the number of matchings of the ladder graph $P_{n+1} \square K_2$ with one pendant leaf attached at every 1-based odd column on a uniform rail (the alternating-attachment variant differs from the uniform-side convention from $n = 3$ on and is a different count).

3 Proof

All four proofs are strong induction on the index with the seed window discharged by computation and the step closed by recurrence unfolding and residue tables (`interval_cases/decide`, `mod_cases`, `omega`); Theorems 2, 3 and 4 first prove a clean main lemma without the residue hypothesis and

then substitute it back. The complete final file is reproduced verbatim below (259 lines, no excerpting); it also ships in this deposit under **proofs/**.

Listing 5: Complete proofs of Theorems 1–4, verbatim lines 1–259 (whole file) of **final/01-pend-proved.lean** (sha256 **be3f31cb**); the smaller monospace width is typesetting only, listing bytes are the frozen file’s own.

```

1  /-
2  AI4Math 流水线 · AI 生成 · 2026-09-27 (终稿生成 2026-09-28)
3  部门: 02 形式化部 (dept-formalize) statement 冻结 + 04 军政部 (dept-prove
4  ) proof body 终稿
5  任务线: mossad-pend (pool-mossad-01, pool Part A: 三线兄弟线之 pend 线)
6  来源卡: tasks/20260927-mossad-pend/card.md (闸门一 2026-09-27 立项)
7  主张定稿: tasks/20260927-mossad-pend/phase0/claims.md (T1 主攻 + T2/T3
8  伴随; T4 备选不进本批)
9  初值数据出处: phase0/counts.txt (双算法互证 PASS) + claims.md 主代理复核
10  订正节
11  (order-8 递推、两四阶子列递推、T1 交织恒等式、T2 样式在  $n \leq 30$  数据窗口
12  全项 PASS);
13  本文件五条 def 的初值/系数/样式逐字照抄 claims.md Lean 草稿 (T3 取订正
14  后形态,
15  格式上仅把每行多分枝改为每行一分枝, 语义不变)。
16
17  口径 (按任务卡锁定): 本文件 (含注释) 只使用「初值 + 整数系数递推定义的纯序
18  列」
19  语言 ( $\mathbb{N} \rightarrow \mathbb{Z}$ ), 命题只谈序列的交织/模余数性质; 组合对象语义一律不进本层。
20  索引口径 (0-based Lean  $\leftrightarrow$  1-based 计数):
21  append n = a(n+1); open k = a(2k+1) (奇位侧); epend k = a(2k+2) (偶位
22  侧)。
23
24  【占位围栏】open k 的数值 = OEIS A386889 (Dresden & Demirkol, 2025-09-04
25  挂名,
26  已占位, 数值核到 k=15); 本线新颖性主张严格限定「完整 order-8 递推 +
27  偶数子列 (epend 侧)」; 禁止任何「新序列」泛写。statement 层仅为纯序列命
28  题,
29  占位状态不改变命题真伪, 仅约束下游报告/论文措辞。
30
31  proof body 由军政部完成; 良定义门由宪兵 gate-batch 集中跑。
32  (来源探针: T1=attempts/r6-t1.lean、T2=attempts/r5-t2.lean、
33  T3o=attempts/r4-t3o.lean、T3e=attempts/r3-step-t3e.lean, 均 EXIT=0 通
34  过。)
```

```

25  -/
26
27  import Mathlib
28
29  /-- **序列 append (主列, order-8 仅偶滞后递推) **。
30  初值 a(0..7) = 3, 10, 46, 141, 660, 2015, 9440, 28814;
31  递推 a(n+8) = 16·a(n+6) - 25·a(n+4) + 10·a(n+2) - a(n) (n ≥ 0)。 -/
32  def append : ℕ → ℤ

```

```

33 | 0 => 3
34 | 1 => 10
35 | 2 => 46
36 | 3 => 141
37 | 4 => 660
38 | 5 => 2015
39 | 6 => 9440
40 | 7 => 28814
41 | n + 8 =>
42   16 * apend (n + 6) - 25 * apend (n + 4) + 10 * apend (n + 2) -
    apend n
43
44 /-- **序列 opend (奇位侧子列, 四阶递推) **.
45 初值  $o(0..3) = 3, 46, 660, 9440$  (= apend 在下标  $0, 2, 4, 6$  处之值;
46 数值与 OEIS A386889 一致——该侧已占位, 见文件头围栏);
47 递推  $o(k+4) = 16 \cdot o(k+3) - 25 \cdot o(k+2) + 10 \cdot o(k+1) - o(k)$  ( $k \geq 0$ )。 -/
48 def opend :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
49 | 0 => 3
50 | 1 => 46
51 | 2 => 660
52 | 3 => 9440
53 | k + 4 =>
54   16 * opend (k + 3) - 25 * opend (k + 2) + 10 * opend (k + 1) -
    opend k
55
56 /-- **序列 epend (偶位侧子列, 四阶递推) **.
57 初值  $e(0..3) = 10, 141, 2015, 28814$  (= apend 在下标  $1, 3, 5, 7$  处之值);
58 递推  $e(k+4) = 16 \cdot e(k+3) - 25 \cdot e(k+2) + 10 \cdot e(k+1) - e(k)$  ( $k \geq 0$ )。 -/
59 def end :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
60 | 0 => 10
61 | 1 => 141
62 | 2 => 2015
63 | 3 => 28814
64 | k + 4 =>
65   16 * end (k + 3) - 25 * end (k + 2) + 10 * end (k + 1) -
    end k
66
67 /-- **T1 (主攻) **: 主列 order-8 递推 = 两条同系数四阶子列的交织——
68 对一切  $k$ ,  $apend(2k) = opend k$  且  $apend(2k+1) = end k$ 。 -/
69 theorem apend_interleave (k :  $\mathbb{N}$ ) :
70   apend (2 * k) = opend k  $\wedge$  apend (2 * k + 1) = end k := by
71   induction k using Nat.strong_induction_on with
72   | h k ih =>
73     by_cases hk : k < 4
74     · interval_cases k <;> decide
75     · obtain ⟨j, rfl⟩ :  $\exists j, k = j + 4$  := ⟨k - 4, by omega⟩
76       constructor
77       · have e8 :  $2 * (j + 4) = (2 * j) + 8$  := by ring
78         rw [e8]

```

```

79     simp only [apend, opend]
80     rw [show 2 * j + 6 = 2 * (j + 3) by ring,
81         show 2 * j + 4 = 2 * (j + 2) by ring,
82         show 2 * j + 2 = 2 * (j + 1) by ring,
83         (ih (j + 3) (by omega)).1, (ih (j + 2) (by omega)).1,
84         (ih (j + 1) (by omega)).1, (ih j (by omega)).1]
85     · have e9 : 2 * (j + 4) + 1 = (2 * j + 1) + 8 := by ring
86       rw [e9]
87       simp only [apend, epend]
88       rw [show 2 * j + 1 + 6 = 2 * (j + 3) + 1 by ring,
89           show 2 * j + 1 + 4 = 2 * (j + 2) + 1 by ring,
90           show 2 * j + 1 + 2 = 2 * (j + 1) + 1 by ring,
91           (ih (j + 3) (by omega)).2, (ih (j + 2) (by omega)).2,
92           (ih (j + 1) (by omega)).2, (ih j (by omega)).2]
93
94 /-- **余数样式 pat2** (12 留数类  $\mapsto$  余数) :
95 0 $\mapsto$ 1, 1 $\mapsto$ 0, 2 $\mapsto$ 0, 3 $\mapsto$ 1, 4 $\mapsto$ 0, 5 $\mapsto$ 1, 6 $\mapsto$ 0, 7 $\mapsto$ 0, 8 $\mapsto$ 1, 9 $\mapsto$ 1, 10 $\mapsto$ 0, 11 $\mapsto$ 1.
96 -/
97 def pat2 :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
98 | 0 => 1
99 | 1 => 0
100 | 2 => 0
101 | 3 => 1
102 | 4 => 0
103 | 5 => 1
104 | 6 => 0
105 | 7 => 0
106 | 8 => 1
107 | 9 => 1
108 | 10 => 0
109 | _ => 1
110
111 /-- **T2 (伴随) **: apend 的 mod 2 余数仅依赖  $n \bmod 12$ , 样式为 pat2. -/
112 theorem apend_mod2_period12 (n r :  $\mathbb{N}$ ) (hr : n % 12 = r) :
113   apend n % 2 = pat2 r := by
114     have main :  $\forall n : \mathbb{N}, \text{apend } n \% 2 = \text{pat2 } (n \% 12) :=$  by
115       intro n
116       induction n using Nat.strong_induction_on with
117       | h n ih =>
118         by_cases hn : n < 8
119         · interval_cases n <;> decide
120         · obtain ⟨j, rfl⟩ :  $\exists j, n = j + 8 :=$  ⟨n - 8, by omega⟩
121           have h1 : apend (j + 8) % 2 = (apend (j + 4) % 2 + apend j % 2) %
122             2 := by
123               simp only [apend]
124               omega
125               rw [h1, ih (j + 4) (by omega), ih j (by omega)]
126               mod_cases hj : j % 12
127               · have hj0 : j % 12 = 0 := by simp [Nat.ModEq] using hj

```

```

126     have h4 : (j + 4) % 12 = 4 := by omega
127     have h8 : (j + 8) % 12 = 8 := by omega
128     rw [hj0, h4, h8]; decide
129   · have hj1 : j % 12 = 1 := by simp [Nat.ModEq] using hj
130     have h4 : (j + 4) % 12 = 5 := by omega
131     have h8 : (j + 8) % 12 = 9 := by omega
132     rw [hj1, h4, h8]; decide
133   · have hj2 : j % 12 = 2 := by simp [Nat.ModEq] using hj
134     have h4 : (j + 4) % 12 = 6 := by omega
135     have h8 : (j + 8) % 12 = 10 := by omega
136     rw [hj2, h4, h8]; decide
137   · have hj3 : j % 12 = 3 := by simp [Nat.ModEq] using hj
138     have h4 : (j + 4) % 12 = 7 := by omega
139     have h8 : (j + 8) % 12 = 11 := by omega
140     rw [hj3, h4, h8]; decide
141   · have hj4 : j % 12 = 4 := by simp [Nat.ModEq] using hj
142     have h4 : (j + 4) % 12 = 8 := by omega
143     have h8 : (j + 8) % 12 = 0 := by omega
144     rw [hj4, h4, h8]; decide
145   · have hj5 : j % 12 = 5 := by simp [Nat.ModEq] using hj
146     have h4 : (j + 4) % 12 = 9 := by omega
147     have h8 : (j + 8) % 12 = 1 := by omega
148     rw [hj5, h4, h8]; decide
149   · have hj6 : j % 12 = 6 := by simp [Nat.ModEq] using hj
150     have h4 : (j + 4) % 12 = 10 := by omega
151     have h8 : (j + 8) % 12 = 2 := by omega
152     rw [hj6, h4, h8]; decide
153   · have hj7 : j % 12 = 7 := by simp [Nat.ModEq] using hj
154     have h4 : (j + 4) % 12 = 11 := by omega
155     have h8 : (j + 8) % 12 = 3 := by omega
156     rw [hj7, h4, h8]; decide
157   · have hj8 : j % 12 = 8 := by simp [Nat.ModEq] using hj
158     have h4 : (j + 4) % 12 = 0 := by omega
159     have h8 : (j + 8) % 12 = 4 := by omega
160     rw [hj8, h4, h8]; decide
161   · have hj9 : j % 12 = 9 := by simp [Nat.ModEq] using hj
162     have h4 : (j + 4) % 12 = 1 := by omega
163     have h8 : (j + 8) % 12 = 5 := by omega
164     rw [hj9, h4, h8]; decide
165   · have hj10 : j % 12 = 10 := by simp [Nat.ModEq] using hj
166     have h4 : (j + 4) % 12 = 2 := by omega
167     have h8 : (j + 8) % 12 = 6 := by omega
168     rw [hj10, h4, h8]; decide
169   · have hj11 : j % 12 = 11 := by simp [Nat.ModEq] using hj
170     have h4 : (j + 4) % 12 = 3 := by omega
171     have h8 : (j + 8) % 12 = 7 := by omega
172     rw [hj11, h4, h8]; decide
173   rw [← hr]; exact main n
174

```

```

175 /-- **余数样式 pat6**:  $0 \mapsto 1, 4 \mapsto 1$ , 其余  $\mapsto 0$  (订正后口径: 奇  $\iff k \equiv 0$  或  $4 \pmod{6}$ )。 -/
176 def pat6 :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
177   | 0 => 1
178   | 4 => 1
179   | _ => 0
180
181 /-- **余数样式 pat3**:  $0 \mapsto 0$ , 其余  $\mapsto 1$  (订正后口径: 偶  $\iff k \equiv 0 \pmod{3}$ )。 -/
182 def pat3 :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
183   | 0 => 0
184   | _ => 1
185
186 /-- **T3 之 opend (伴随) **: opend  $k$  的  $\text{mod } 2$  余数仅依赖  $k \text{ mod } 6$ , 样式为
    pat6
    (即 opend  $k$  为奇  $\iff k \equiv 0$  或  $4 \pmod{6}$ )。 -/
187 theorem opend_mod2_period6 (k r :  $\mathbb{N}$ ) (hr :  $k \% 6 = r$ ) :
188   opend  $k \% 2 = \text{pat6 } r := \text{by}$ 
189   have main :  $\forall k : \mathbb{N}, \text{opend } k \% 2 = \text{pat6 } (k \% 6) := \text{by}$ 
190     intro k
191     induction k using Nat.strong_induction_on with
192     | h k ih =>
193       by_cases hk :  $k < 4$ 
194       · interval_cases k <;> decide
195       · obtain ⟨j, rfl⟩ :  $\exists j, k = j + 4 := \langle k - 4, \text{by omega} \rangle$ 
196         have h1 :  $\text{opend } (j + 4) \% 2 = (\text{opend } (j + 2) \% 2 + \text{opend } j \% 2) \% 2 := \text{by}$ 
197           simp only [opend]
198           omega
199         rw [h1, ih (j + 2) (by omega), ih j (by omega)]
200         mod_cases hj :  $j \% 6$ 
201         · have hj0 :  $j \% 6 = 0 := \text{by simp} [\text{Nat.ModEq}] \text{ using } hj$ 
202           have h2 :  $(j + 2) \% 6 = 2 := \text{by omega}$ 
203           have h4 :  $(j + 4) \% 6 = 4 := \text{by omega}$ 
204           rw [hj0, h2, h4]; decide
205         · have hj1 :  $j \% 6 = 1 := \text{by simp} [\text{Nat.ModEq}] \text{ using } hj$ 
206           have h2 :  $(j + 2) \% 6 = 3 := \text{by omega}$ 
207           have h4 :  $(j + 4) \% 6 = 5 := \text{by omega}$ 
208           rw [hj1, h2, h4]; decide
209         · have hj2 :  $j \% 6 = 2 := \text{by simp} [\text{Nat.ModEq}] \text{ using } hj$ 
210           have h2 :  $(j + 2) \% 6 = 4 := \text{by omega}$ 
211           have h4 :  $(j + 4) \% 6 = 0 := \text{by omega}$ 
212           rw [hj2, h2, h4]; decide
213         · have hj3 :  $j \% 6 = 3 := \text{by simp} [\text{Nat.ModEq}] \text{ using } hj$ 
214           have h2 :  $(j + 2) \% 6 = 5 := \text{by omega}$ 
215           have h4 :  $(j + 4) \% 6 = 1 := \text{by omega}$ 
216           rw [hj3, h2, h4]; decide
217         · have hj4 :  $j \% 6 = 4 := \text{by simp} [\text{Nat.ModEq}] \text{ using } hj$ 
218           have h2 :  $(j + 2) \% 6 = 0 := \text{by omega}$ 
219

```

```

220         have h4 : (j + 4) % 6 = 2 := by omega
221         rw [hj4, h2, h4]; decide
222         · have hj5 : j % 6 = 5 := by simp [Nat.ModEq] using hj
223         have h2 : (j + 2) % 6 = 1 := by omega
224         have h4 : (j + 4) % 6 = 3 := by omega
225         rw [hj5, h2, h4]; decide
226     rw [← hr]; exact main k
227
228 /-- **T3 之 epend (伴随) **: epend  $k$  的  $\text{mod } 2$  余数仅依赖  $k \text{ mod } 3$ , 样式为
    pat3
    (即 epend  $k$  为偶  $\iff k \equiv 0 \pmod{3}$ ) )。 -/
229 theorem endep_mod2_period3 (k r : ℕ) (hr : k % 3 = r) :
230     endep k % 2 = pat3 r := by
231         intro k
232         induction k using Nat.strong_induction_on with
233         | h k ih =>
234             by_cases hk : k < 4
235             · interval_cases k <|> decide
236             · obtain ⟨j, rfl⟩ : ∃ j, k = j + 4 := ⟨k - 4, by omega⟩
237                 have h1 : endep (j + 4) % 2 = (endep (j + 2) % 2 + endep j % 2) %
238                     2 := by
239                         simp only [epend]
240                         omega
241                 rw [h1, ih (j + 2) (by omega), ih j (by omega)]
242                 mod_cases hj : j % 3
243                 · have hj0 : j % 3 = 0 := by simp [Nat.ModEq] using hj
244                 have h2 : (j + 2) % 3 = 2 := by omega
245                 have h4 : (j + 4) % 3 = 1 := by omega
246                 rw [hj0, h2, h4]
247                 decide
248                 · have hj1 : j % 3 = 1 := by simp [Nat.ModEq] using hj
249                 have h2 : (j + 2) % 3 = 0 := by omega
250                 have h4 : (j + 4) % 3 = 2 := by omega
251                 rw [hj1, h2, h4]
252                 decide
253                 · have hj2 : j % 3 = 2 := by simp [Nat.ModEq] using hj
254                 have h2 : (j + 2) % 3 = 1 := by omega
255                 have h4 : (j + 4) % 3 = 0 := by omega
256                 rw [hj2, h2, h4]
257                 decide
258     rw [← hr]; exact main k
259

```

4 Verification evidence

- Toolchain: Lean v4.34.0 (leanprover/lean4:v4.34.0), mathlib4 v4.34.0, revision 5ed29652. Reproduction: with that pin, lake env

`lean` on `proofs/01-pend-proved.lean` compiles the file with exit code 0 and empty output (measured 70.0s in the audit’s independent strict re-run).

- Statement integrity: the frozen statement file `formalized/01-pend-statements.lean` (113 lines) has sha256

```
78bccc97afba79adea9658a2158d17b2
ed247b584d9f28266e6fc7914ba9f922,
```

and the final file `final/01-pend-proved.lean` (259 lines) has sha256

```
be3f31cb365aa7b805dce0d29e562b5d
fdb12617007f888e67564dacfd22540b.
```

The audit extracted the ten declaration blocks (six definitions, four theorems, docstrings included) from both files programmatically: 10/10 byte-identical; a whole-file diff shows exactly six hunks, namely three header-comment lines and the four `sorry` placeholder lines replaced by proof bodies – the statement body itself has zero drift.

- Kernel check: compiles with 0 `sorry` (whole-file scan including comments, case-insensitive, also covering `admit`, `sorryAx`, `native_decide`, `axiom` and `unsafe`, all zero hits); `#print axioms` output, verbatim (kept literal; the smaller monospace width is typesetting only):

```
1 'apend_interleave' depends on axioms: [propext, Quot.sound]
2 'apend_mod2_period12' depends on axioms: [propext, Quot.sound]
3 'opend_mod2_period6' depends on axioms: [propext, Quot.sound]
4 'epend_mod2_period3' depends on axioms: [propext, Quot.sound]
```

Each line is a subset of the whitelist `{propext, Quot.sound, Classical.choice}` – in fact `Classical.choice` is not even used. The strict re-compile emits no error and no warning at all (stdout empty); the audit’s independent re-runs (70.0s strict, 41.3s axioms) are byte-identical to the self-reported outputs. The evidence ships as `audit/audit-strict-compile.log` and `audit/audit-axioms-sidecar.log` (the four lines above are that log’s payload; the shipped copy adds a three-line provenance header, and the audit’s re-run `audit/axioms-recheck.out` is byte-identical to the task’s own axiom printout).

- Grading by the audit department: all four theorems *complete proof*; lowest grade reached for the case = *complete proof*; no scaffolding route was used (the proof skeleton replicates a same-pool kernel-verified precedent, which the audit ruled is not a scaffolding downgrade).
- Audit chain (originals in this deposit under `audit/`): final adjudication `final-01-pend.txt` (eight independent checks), well-definedness verdict `welldef-verdict.md` with gate logs `gate-t1.txt`, `gate-t2.txt`, `gate-t3o.txt`, `gate-t3e.txt` and the supplementary ring probe `gate-t1-ring.txt` (degeneracy probes 33/33 FAIL, i.e. no one-tactic shortcut), the statement symmetric-difference record `statement-diff.out`, the audit’s re-verification `axioms-recheck.out`, and this note’s claim check `claim-check.md` — added to `audit/` after the audit department returns its adjudication, and a prerequisite for publication. The semantic-fidelity check (roundtrip re-translation) of the underlying task ran on the same model node as its formalisation; that independence weakness is mitigated, not removed, by kernel adjudication and by two independent audit re-runs.

5 Novelty evidence

Adjudication copied from the pipeline’s exclusivity review (C9 gate), which ran before the problem was accepted and re-derived the data independently: verdict **no occupying record found for the main sequence; the odd-position subsequence is already occupied (OEIS A386889), value tier new sequence / new recurrence (with an occupancy warning)**. It is not a claim of new mathematics and not a first-discovery claim, and no “new sequence” claim about the whole family is made. Summary by channel; the full query log, including every zero-hit query, is in this deposit at `audit/c9-record.md`.

channel	queries	result
OEIS main-sequence variants [5]	11	zero hits: original string, drop-1/2/3 (4), -1, +1, -1 then drop-1, +1 then drop-1 (4), -2 (1), ×2, ×2 then drop-1 (2); ÷2 not applicable (first term odd), recorded
OEIS odd-position subsequence	6	A386889 [6] on the plain string and two reformulations; -1, +1, ×2 variants zero hits
OEIS even-position subsequence	6	zero hits (plain, drop-1, 8-term window, -1, +1, ×2)
OEIS keywords	4	unrelated noise only (A000045, A011973)
abMATH	6	two clean zero readings for pendant-ladder matching queries; Farrell 1978 <i>Matchings in ladders</i> and others concern plain ladders, no pendant structures
OpenAlex	3	235 full-text-noise hits; 0 hits; one query HTTP 429, recorded as unverified
arXiv	5	three exact-phrase zeros; author searches found no tiling paper behind A386889
Math Stack Exchange	1	zero items
general web-layer	13 attempts	all unverified (channel failure, recorded as such, not used as zero-literature evidence)
library layer	3	mathlib has matching predicates but no matching-count theorem; compiles none; sequencelib none
public Lean ecosystem	4	repository searches and one full-tree scan: zero hits

Three points of the record are part of the evidence. First, the review’s anchor self-proof (Fibonacci and a known tatami sequence returned their

correct OEIS entries) fixes what “zero hit” means on the sequence channel. Second, the occupying entry: OEIS A386889 (Dresden and Demirkol, 2025-09-04) counts $1 \times 1/1 \times 2$ tilings of a $3 \times (2n - 1)$ strip with every other top-row cell removed; the review aligned 13 terms of the odd-position subsequence against it offset-for-offset and independently verified the graph isomorphism with the pendant ladder (degree sequences and an explicit bijection at $n = 2$), and its recurrence signature $(16, -25, 10, -1)$ equals the one fitted here — so the odd-position side is *occupied* and carries the fence stated in Section 2. Third, the externally supplied review had queried only six main-sequence variants and missed exactly this derived occupancy; the degradation notes (OpenAlex 429, general web layer unavailable, no Chinese-language database searched) are on file. If any channel later returns a hit on the remaining novelty (full order-8 recurrence, even-position subsequence, interleaving, mod-2 patterns), the tier above weakens accordingly, and this deposit will be amended by a later version rather than by editing this record.

6 Scope and limitations

Grade and boundary, stated as the audit states them. (a) The theorems cover the recurrence-defined sequences only; their equality with pendant-ladder matching counts is not kernel-proved and is Conjecture 1. Its support is OEIS/literature endorsement plus local numeric probes: exact counting from the graph definition by three independent methods agreeing on the seed window and extended to $n \leq 30$; the odd-position side was aligned with A386889 over 13 terms and cross-checked to $k = 15$; the well-definedness audit independently recomputed all statements on the window $n \leq 30 / k \leq 14$. The enumerators themselves are not verified, and the scripts and their outputs stay in the working repository under `tasks/20260927-mossad-pend/`. (b) Theorems 3 and 4 are stated in their corrected forms (odd iff $k \equiv 0, 4 \pmod{6}$; even iff $k \equiv 0 \pmod{3}$); the earlier off-by-one formulations ($k \equiv 1, 5; k \equiv 1$) are deprecated and must not be cited. (c) Indexing is 0-based: **apend** $n = a(n + 1)$, **opend** $k = a(2k + 1)$, **epend** $k = a(2k + 2)$. (d) The recurrences are computational findings: Berlekamp–Massey fits over $\mathbb{Q}$ verified digit-for-digit against enumeration and independently recomputed, not derived from a transfer-matrix argument. The older “alternating attachment is robust” belief is refuted: alternating and uniform-side attachments differ from $n = 3$ on $(3, 10, 47, 143, \dots)$ vs $3, 10, 46, 141, \dots$, three methods agreeing), so the combinatorial reading is fixed to the uniform-side convention. (e) The word *minimal* in “minimal period 12/6/3” is supported

by the data window, the audit’s independent recomputation ruling out the candidate divisors, and a $\text{GF}(2)$ factorisation cross-check; the kernel proves the universal statements themselves but not minimality. (f) The gate-four sign-off of the underlying task was exercised by the author’s instruction to deposit this claim on 2026-09-28, while the task report file still carries its draft header; both facts are recorded here as they stand. (g) A mod-4 pattern and arbitrary-modulus period questions were left as an unattempted option and are not implied anywhere. (h) This note carries no literature review and is a deposit record, not a survey; the narrative paper for the same result is separate.

Data availability

The deposit package for this claim contains the claim note (LaTeX source and PDF), the Lean sources (`proofs/`: frozen statement file, final proof file, `lean-toolchain`), the verification and audit logs, and the full C9 query log with zero-hit entries (`audit/`). Zenodo: the DOI is reserved at upload by the author and is recorded in `metadata/README.md` of this package once created; see `UPLOAD.md` in the source repository. Claim note under CC BY 4.0, code under MIT.

Use of generative AI

(a) AI performed: candidate generation and the exclusivity review, auto-formalisation of the six definitions and four statements, the roundtrip semantic check, proof search and proof-script writing, error classification and repair, and the assembly of this note including its verbatim listings. (b) Model, node and budget: all task-stage sampling ran on one node, wire-id `anthropic/a6api-main/kimi-k3`, with no cross-node switching, and consumed 2 LLM calls of a permitted 8 (roundtrip adjudication: one call truncated at the token cap and retried on the same node, one succeeded) and 20 kernel compiles of a permitted 24, copied verbatim from the task ledger. This note consumed 0 pipeline LLM calls against a cap of 4; drafting was mechanical assembly. (c) Final adjudication: all statements and proofs reported here were checked by the Lean 4 kernel: every proof compiles with no `sorry`, and `#print axioms` reports only `{propext, Quot.sound}`. (d) The AI system is not an author; the human author reviewed the evidence and is responsible for all content.

Author contributions

CRediT-style: the human author adjudicated topic selection, statement semantics and the deposit decision; the AI4Math pipeline (an LLM) generated the candidate propositions, the proof searches and the text drafts. The AI system is not listed as an author.

References

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